

OAK-D Pro Object Detection with Distance Measurement

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Overview

This guide explains how to install, configure, and run the OAK-D Pro object detection programs located in the Oak_D_Pro_Config folder of the following repository:

<https://github.com/LiDRON7/Sensor-Control-Data-Gathering>

The repository contains two OAK-D Pro programs:

`Object_Detection.py`

`Object_Detectiton_With_Distance.py`

The user can choose which program to run depending on the desired functionality.

This project uses a Raspberry Pi using Ubuntu together with a Luxonis OAK-D Pro camera.

Object_Detection.py

This program opens the OAK-D Pro camera, detects supported objects, and draws a bounding box around detected objects.

Object_Detection_With_Distance.py

This program opens the OAK-D Pro camera, detects supported objects, draws a bounding box around detected objects, measures the distance between the camera and the detected object, displays the distance in both feet and meters.

Both programs intentionally do not display:

- Object class names
- Object labels
- Detection confidence percentages

Hardware Requirements

The following hardware is required:

- Raspberry Pi
- Ubuntu installed on the Raspberry Pi
- Luxonis OAK-D Pro
- Proper USB cable for the OAK-D Pro
- Adequate power supply for the Raspberry Pi
- Monitor or graphical display if using `cv2.imshow()`

A **high-quality Raspberry Pi power supply is required** because the OAK-D Pro can draw significant power through USB.

Software Requirements

The programs require:

- Python3
- Python virtual environments
- DepthAI v3
- OpenCV
- NumPy

The `depthai-core` repository is **not** required for the program to run. It is useful mainly for:

- Reviewing official Luxonis examples
- Learning the DepthAI API
- Testing different camera examples
- Developing DepthAI itself

DepthAI is installed as a Python package inside a virtual environment.

Updating Ubuntu and Installing Required System Packages

Enter Terminal on the Raspberry Pi and update the package information:

```
Shell
sudo apt update
```

Then install available OS updates:

```
Shell
sudo apt upgrade -y
```

Install Python, Git, virtual-environment support, and the required USB libraries:

```
Shell
sudo apt install -y \
python3 \
python3-pip \
python3-venv \
git \
libusb-1.0-0-dev \
libudev-dev
```

These packages provide the basic tools required to run the project.

Clone the GitHub Repository

Go to the home directory

```
Shell
cd ~
```

Clone the repository:

```
Shell
git clone https://github.com/LiDRON7/Sensor-Control-Data-Gathering.git
```

Enter the repository:

```
Shell
cd ~/Sensor-Control-Data-Gathering
```

Then enter the OAK-D Pro configuration folder:

```
Shell
cd Oak_D_Pro_Config
```

The terminal should now be inside:

```
~/Sensor-Control-Data-Gathering/Oak_D_Pro_Config
```

You can verify the files using:

```
Shell  
ls
```

You should see files including:

```
Object_Detection.py  
Object_Detection_With_Distance.py  
install_requirements.txt
```

Create a Python Virtual Environment

While inside `~/Sensor-Control-Data-Gathering/Oak_D_Pro_Config`, create a virtual environment:

```
Shell  
python3 -m venv .venv
```

This creates a new `.venv/` file inside the folder. The `.venv` directory contains an isolated Python environment for the project. This prevents the project from modifying Ubuntu's system Python installation.

Activate the Virtual Environment

Activate the environment:

```
Shell  
source .venv/bin/activate
```

The terminal should now show something similar to:

```
(.venv) user@raspberrypi:~/Sensor-Control-Data-Gathering/Oak_D_Pro_Config$
```

The (.venv) at the beginning indicates that the virtual environment is active.

Upgrade pip

Upgrade pip inside the virtual environment:

```
Shell
python3 -m pip install --upgrade pip
```

Install the Python Requirements

The install_requirements.txt file contains the Python packages required by the OAK-D Pro programs. Install them with:

```
Shell
python3 -m pip install -r install_requirements.txt
```

The requirements file contains:

```
depthai>=3.0.0,<4.0.0
opencv-python>=4.8.0
numpy>=1.24.0
```

This installs DepthAI v3 and OpenCV inside the virtual environment.

Why a Virtual Environment is Used

Modern Ubuntu versions protect the system Python installation. Attempting to install DepthAI outside a virtual environment may produce externally-managed-environment. This is expected behavior on newer Ubuntu versions. Always use source .venv/bin/activate before installing or running the program.

Verify DepthAI, OpenCV, and NumPy

Check that DepthAI and OpenCV is installed:

Shell

```
python -c "import depthai as dai; import cv2; print('DepthAI:', dai.__version__);  
print('OpenCV:', cv2.__version__); print('NumPy:', np.__version__)"
```

A successful result should look similar to:

DepthAI: 3.x.x
OpenCV: 4.x.x
NumPy: x.x.x

At this point, the Python environment is ready.

Choose Which Program to Run

There are two programs available.

Option 1 – Object Detection Only

Use `Object_Detection.py`. This option should be used when only object locations are needed.

Run it with:

Shell

```
python3 Object_Detection.py
```

The camera window will display bounding boxes around detected objects.

Option 2 – Object Detection with Distance

Use `Object_Detection_With_Distance.py`. This option should be used when object locations and distance are needed.

Run it with:

Shell

```
python3 Object_Detection_With_Distance.py
```

This program performs object detection and stereo-depth measurement. Each detected object will have a bounding box, distance in meters, and distance in feet.

If you run into an error

If you run into this error:

```
Shell
Error: [depthai] [warning] Insufficient permissions to communicate with
X_LINK_UNBOOTED device with name "7.1". Make sure udev rules are set
Traceback (most recent call last):
  File
"/home/lidron/Documents/oakd-pro-ws/Sensor-Control-Data-Gathering/Oak_D_Pro_Config
/Object_Detection.py", line 18, in <module>
    with dai.Pipeline() as pipeline:
        ^^^^^^^^^^^^^^^^^
RuntimeError: No available devices
```

Run this:

```
Shell
echo 'SUBSYSTEM=="usb", ATTRS{idVendor}=="03e7", MODE="0666"' | sudo tee
/etc/udev/rules.d/80-movidius.rules && sudo udevadm control --reload-rules && sudo
udevadm trigger
```

Where the output is:

```
Shell
SUBSYSTEM=="usb", ATTRS{idVendor}=="03e7", MODE="0666"
```

This will reload and activate the new rules immediately without needing to restart the computer.

Closing the Program

When the OpenCV camera window is active, press `q` to close the program. After finishing, the virtual environment can be disabled with `deactivate`.

Running the Program Again Later

The dependencies only need to be installed once. After restarting the Raspberry Pi, navigate back to the project:

```
Shell
cd ~/Sensor-Control-Data-Gathering/Oak_D_Pro_Config
```

Activate the virtual environment:

```
Shell
source .venv/bin/activate
```

Then choose one of the programs (see [Choose Which Program to Run](#)).

Contact

Any questions feel free to ask 😊

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